

backend/core/training.py Documentation (Roman Urdu)

Yeh AI system ka "Extraction aur Formatting" hissa hy. Jab user apney local machine me SoloLLM AI Model ko "Self Train (Fine Tune)" karna chahey ga tou usko lakho'n purani chats or text ka data sahi AI understandable format m chaye hoga, yeh script wo banati ha!

File Ka Maqsad (Purpose)

Is file ka bunyadi maqsad Local Database (SQLite) sy Purani Chats (Conversations) aur Pata ki gai Docs (workspace PDFs wagera) se Text block ukhadna aur usmein user ki prompt aur assistant k jawab match bana kr instruction format (HuggingFace trl) bnana a.k.a JsonL formatting krna!

Code Blocks aur Classes ki Tafseel

1. class TrainingProgress / TrainingConfig

- **Data Models:** Fine tuning parameters default (Learning Rate $2e-4$, num_epochs 3, Wagira), aur Training Progress Bar status ko update karny k variables is me set kye jtay hein.

2. class TrainingDataPreparer

- Yahi wo main engine class h.
- **_dedupe_examples:** Training mein copy-paste duplication models ko tbah krdeta hai, yeh math hashes bna k similar lines discard (deduplicate) kardeta ha.
- **extract_from_documents():** Database me saved chunk blocks sy line by line data parta hai. Chunk level text p random generate kr k instruction parameters set krdta hai *"Summarize the key points from file..."* aur document excerpt ko Output lagakar system ko training datadet tyar kardeta.
- **extract_from_conversations():** Messages table parse kr k `user` vs `assistant` ka data prepare krta hv, sth k history context messages check krta ha. Aik smart Quality Score Filter lgia hua ha *"Assistant message minimum 20 chars hna zruri tabhi munasib training log bnayga"*.

- **split_train_validation()**: Poora dataset ko standard machine learning principle (Validation Split) se Random Shuffle maarta hai k 90% data train mein jaye ga ar 10% valid/loss chk k liy use houga.
- **format_for_training()**: Har entry ko specific chat-template array structure (`[{"role": "user", "content": ...}]`) list dict format m generate kre dy dtay ta ky seedha Lora configuration mein pass ki jyeskyn.